



# RansoTrail: Early Ransomware Detection System

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## Project Documentation

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# Summary

Cybersecurity threats continue to evolve, with ransomware remaining one of the most disruptive forms of cybercrime affecting organizations worldwide. Modern ransomware attacks can result in financial losses, operational downtime, reputational damage, regulatory penalties, and data loss. As a result, organizations require proactive security solutions capable of identifying malicious behavior before a large-scale attack occurs.

Ransotrail was developed as a ransomware detection, correlation, and prediction platform designed to monitor suspicious system activity, identify potential ransomware behavior, assess threat severity, estimate potential impact, and provide security analysts with actionable intelligence through reporting and visualization.

The platform combines multiple monitoring components, including file activity monitoring, PowerShell monitoring, registry monitoring, and privilege escalation detection. These components feed into a centralized threat correlation engine that calculates risk scores, assigns threat levels, predicts attack progression, estimates financial impact, generates incident reports, and supports dashboard-based threat visualization.

This project demonstrates how behavioral analysis and threat correlation can be used to detect early warning signs of ransomware attacks before significant damage occurs.

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## Project Objectives

The primary objectives of the project were:

- Develop a file activity monitoring system capable of detecting suspicious file behavior like deletion, locking and modifications of files.
- Detects PowerShell activity commonly associated with malicious execution.
- Monitor Windows Registry changes that may indicate persistence mechanisms.
- Detect privilege escalation events.
- Correlate security events from multiple sources using a correlation engine.
- Calculate ransomware risk scores.
- Classify threats based on severity levels.
- Estimate potential attack progression.
- Estimate potential financial impact.

- Generate automated incident reports.
  - Record threat intelligence logs in a csv file.
  - Visualize security events through an interactive power bi dashboard.
  - Demonstrate proactive ransomware detection techniques.
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## Project Architecture

The Correlock architecture consists of several integrated components:

File Monitor



PowerShell Monitor



Registry Monitor



Privilege Escalation Monitor



Threat Correlation Engine



Risk Scoring System



Prediction Engine



Incident Reporting



Threat Logging



# Development Phases

## Phase 1: File Activity Monitoring

The first phase focused on monitoring file system activity.

The file monitor was developed using Python and the Watchdog library to track:

- File Creation
- File Modification
- File Deletion
- File Renaming

The monitor records events in real time and assigns risk scores based on the type of activity observed.

Risk allocation included:

- Created File = 2 points
- Modified File = 5 points
- Renamed File = 8 points
- Deleted File = 10 points

The monitor also records activity into CSV log files for later analysis which was used to create a csv file.

## Reason

Ransomware frequently performs mass file modifications, renaming, and deletion operations. Monitoring these activities provides an early indicator of potential ransomware behavior.

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## Phase 2: PowerShell Monitoring

The second phase involved monitoring PowerShell execution.

PowerShell is commonly used by attackers because it allows:

- Malware execution
- Script-based attacks
- Lateral movement
- Persistence creation

Detection of PowerShell activity contributed additional risk within the threat correlation process.

### Reason

PowerShell is one of the most commonly used tools during ransomware attacks and therefore represents a valuable threat indicator.

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## Phase 3: Registry Monitoring

A registry monitor was developed using the Windows Registry API.

The monitor observes changes within the Windows Registry and identifies newly created or modified entries.

### Reason

Attackers often modify registry keys to:

- Maintain persistence
- Execute malware at startup
- Disable security controls
- Modify system configurations

Registry modifications therefore serve as important indicators of malicious behavior.

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## Phase 4: Privilege Escalation Monitoring

A privilege escalation monitor was implemented using Windows administrative privilege checks.

The monitor detects transitions from standard user privileges to administrator privileges.

### Reason

Privilege escalation is a critical stage in many cyberattacks because it allows attackers to gain broader control over systems and data.

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## Phase 5: Threat Correlation Engine

One of the most important components of the project was the correlation engine.

Rather than evaluating security events independently, the engine combines indicators from multiple monitoring sources so that they can work as a one source.

Examples include:

- High file activity + PowerShell execution
- Registry changes + privilege escalation
- File encryption behavior + persistence activity

This approach increases confidence in detections and reduces false positives.

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## Phase 6: Risk Scoring System

The risk scoring system quantifies threat severity.

Indicators contributing to risk scores include:

- File Events
- PowerShell Activity
- Registry Changes
- Privilege Escalation

Threat levels are classified as:

LOW

MEDIUM

HIGH

CRITICAL

The classification system allows analysts to prioritize incidents effectively.

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## **Phase 7: Prediction Engine**

The prediction engine introduced proactive security capabilities.

### **Time-to-Compromise Estimation**

The system estimates how quickly a ransomware attack could compromise a target environment based on observed activity rates.

#### **Reason**

Provides organizations with an understanding of attack progression speed and urgency.

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## **Phase 8: Financial Impact Estimation**

Technical security findings were translated into estimated financial consequences.

Estimated values:

LOW → R1,000

MEDIUM → R10,000

HIGH → R50,000

CRITICAL → R250,000

## **Purpose**

Allows technical findings to be communicated in business terms.

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## **Phase 9: Incident Report Generator**

An automated reporting component was implemented.

Reports contain:

- Threat Level
- Risk Score
- File Events
- Registry Changes
- PowerShell Activity
- Privilege Escalation
- Time-to-Compromise Estimate
- Financial Impact Estimate

## **Purpose**

Provides structured documentation for incident response and investigations.

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## **Phase 10: Threat Logging**

Threat logging was implemented to create historical records of security events.

Threat logs support:

- Trend analysis
- Security reporting
- Dashboard visualization
- Incident investigations
- Compliance activities

All threat logs are generated in a csv file to conduct data analytics on the data.

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# Testing and Validation

The project was tested using simulated ransomware-like activity.

Testing scenarios included:

## **Mass File Creation**

Large numbers of files were created to simulate attacker behavior.

## **Mass File Modification**

Files were modified repeatedly to simulate encryption activity.

## **Mass File Renaming**

Files were renamed to simulate ransomware changing file extensions.

## **Mass File Deletion**

Files were deleted to simulate destructive behavior.

## **PowerShell Activity**

PowerShell execution was simulated to test detection capabilities.

## **Registry Changes**

Registry keys were added and modified to test persistence detection.

## **Privilege Escalation**

Administrative privileges were simulated to test escalation detection.

Results demonstrated successful event detection and threat classification.

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# Dashboard Development

A Power BI dashboard was developed to visualize threat intelligence generated by the platform.

Dashboard components included:

- Total Threats
- Average Risk Score
- Critical Threat Count
- PowerShell Detection Count
- Threat Level Distribution
- Risk Score Timeline
- Registry Activity Analysis
- Privilege Escalation Analysis
- High-Risk Incident Table

## **Purpose**

The dashboard converts raw security data into actionable intelligence for analysts and decision-makers.

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# **Message to Companies and Organizations**

The primary message of this project is that ransomware attacks rarely occur without warning.

Attackers often generate observable indicators before encryption begins, including:

- Suspicious file activity
- PowerShell execution
- Registry modifications
- Privilege escalation attempts

Organizations that monitor and correlate these indicators can detect threats earlier and prevent larger incidents.

Early detection enables:

- Faster response
  - Reduced downtime
  - Lower recovery costs
  - Reduced financial losses
  - Improved business continuity
  - Stronger cybersecurity resilience
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## Business Value

RansoTrail provides value through:

- Early Threat Detection
  - Risk-Based Prioritization
  - Security Event Correlation
  - Threat Visualization
  - Incident Reporting
  - Business Impact Awareness
  - Improved Decision Making
  - Enhanced Security Posture
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## Future Improvements

Future enhancements may include:

- Machine Learning Models
  - SIEM Integration
  - Email Alerting
  - Real-Time Notifications
  - Cloud Monitoring
  - MITRE ATT&CK Mapping
  - Threat Intelligence Feeds
  - Automated Incident Response
  - Threat Logging
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# Conclusion

RansoTrail successfully demonstrates how multiple cybersecurity capabilities can be integrated into a unified ransomware detection and prediction platform.

Through behavioral monitoring, event correlation, risk scoring, prediction, reporting, threat logging, and visualization, the platform provides organizations with the ability to identify threats before significant damage occurs.

The project highlights the importance of proactive cybersecurity monitoring and illustrates how early warning indicators can be transformed into actionable intelligence that supports both security operations and business decision-making, saving organizations from large-scale attacks.

Ultimately, RansoTrail demonstrates that effective ransomware defense requires not only detection, but also correlation, prediction, reporting, and continuous visibility into the security posture of an organization.